

Network Devices

What is a network?

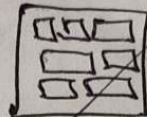
→ A computer network is a digital telecommunication network which allows nodes to share resources.



router



switch

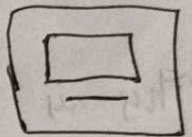


fire wall



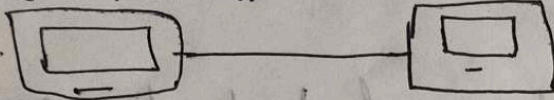
server

end hosts
end points



client

Building network



pc1

pc2

→ two pc's connected together actually makes network

clients



→ A client is a device that accesses a service made available by server

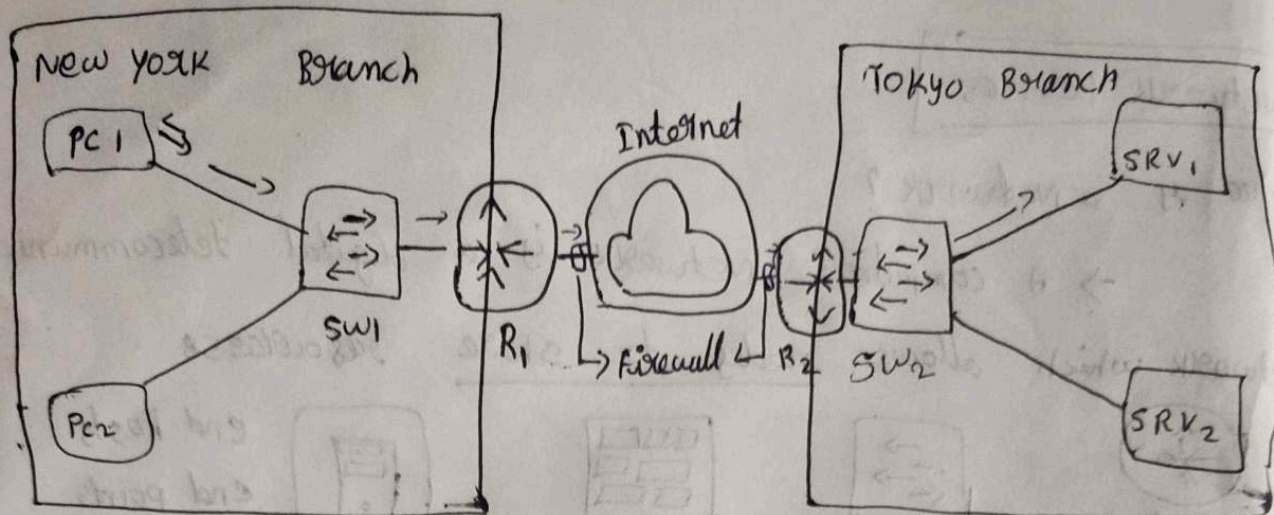
server



→ a device that provides functions or services for clients.

*The same device can be a client in some situations, and a server in other situations.

Switches - catalyst 9200 & 3650



- This is next part of the connections between end host & Internet
- Typically you don't connect end hosts like PCs or servers directly to each other.
- Aggregate the connections to device called Switch.
- However, these switches cannot connect directly to the Internet.
- have many network interfaces / ports for end hosts to connect to (usually 24+).
- Provide connectivity to hosts within the same LAN
- do not provide connectivity between LANs / over the Internet.

To do so we need another kind of network device.

↳ Router

Routers $\Rightarrow$ ISR 1000, ISR 900, ISR 4000

- $\rightarrow$ have fewer network interfaces than switches.
- $\rightarrow$ are used to provide connectivity between LANs.
- $\rightarrow$ are therefore used to send data over the Internet.

Firewalls $\Rightarrow$ ASA 5500-K | Fire power 2100

$\Rightarrow$ monitor & control network traffic based on configured rules.

- $\rightarrow$ you explicitly configure which network traffic should be allowed in to your network & which should not?

$\Rightarrow$ Can be placed inside the network, or outside the network.

- $\rightarrow$ firewall can filter traffic before it reaches the router or after it has passed through the router.

$\Rightarrow$ are known as 'next-generation firewalls' when they include more modern & advanced filtering capabilities.

what about the firewall on your computer?

network firewalls..

are hardware devices that filter traffic between networks.

Host-based firewalls..

are software applications that filter traffic entering & exiting a host machine, like a PC